Questions

Q1.

The speed of light is 3.0×10^8 m/s.

The wavelength of yellow light is 5.8×10^{-7} m.

Calculate the frequency of yellow light.

State the unit.

Use the equation

$$\text{frequency} = \frac{\text{speed}}{\text{wavelength}}$$

(3)

frequency = ... unit .....

(Total for question = 3 marks)

Q2.

Figure 11 shows a large tank of water.

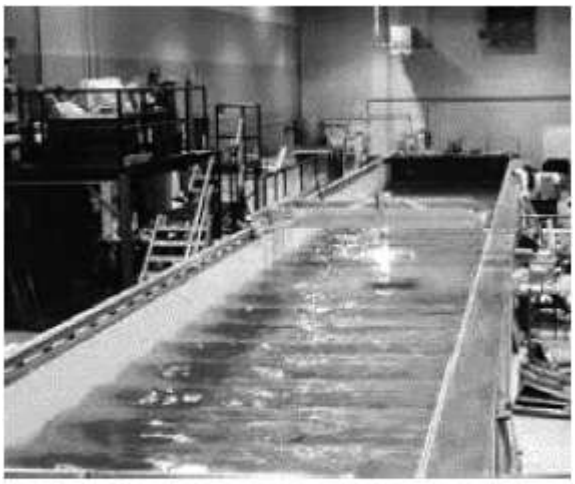

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Figure 11

The tank of water is used to study water waves.

Figure 13 shows part of the inside of the Earth below the surface.

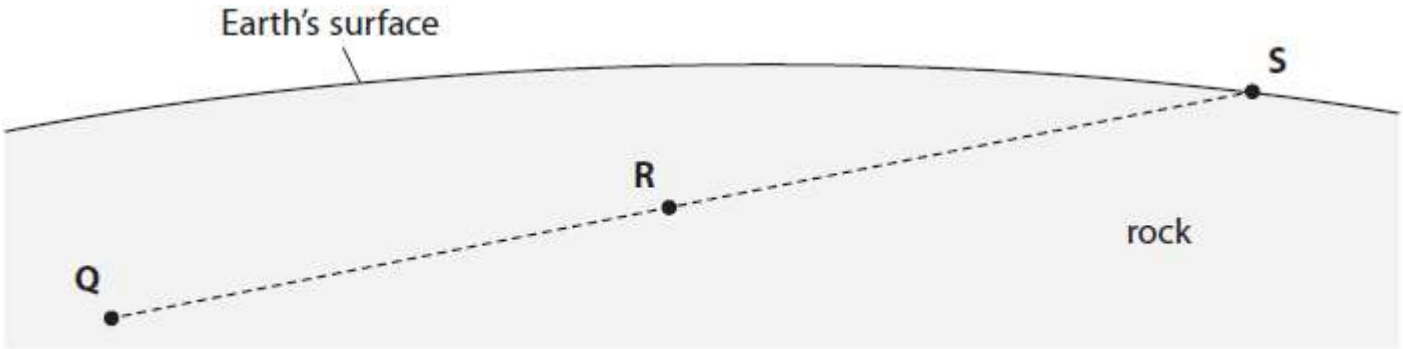

Figure 13

An earthquake starts at **Q**.

A seismic wave travels from **Q** to **S**.

The seismic wave is a longitudinal wave.

The frequency of the seismic wave is 12Hz.

A technician measured the frequency of the water wave in Figure 11 by counting how many waves passed him in 15 s.

Explain why this would **not** be a suitable method for measuring the frequency of the seismic wave in Figure 13.

(2)

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(Total for question = 2 marks)

Q3.

(a) Seismic (earthquake) waves can be either P-waves or S-waves.

Which row of the table is correct for P-waves?
Put a cross (☒) in the box next to your answer.

(1)

	type of wave	can they be refracted?
☐ A	longitudinal	yes
☐ B	transverse	no
☐ C	longitudinal	no
☐ D	transverse	yes

(b) Explain why it is difficult to predict when an earthquake will happen.

(2)

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Q4.

Sound waves are longitudinal waves.

Water waves are transverse waves.

Describe the difference between longitudinal waves and transverse waves.

(3)

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(Total for question = 3 marks)

Q5.

Figure 16 is a diagram of part of a water wave.

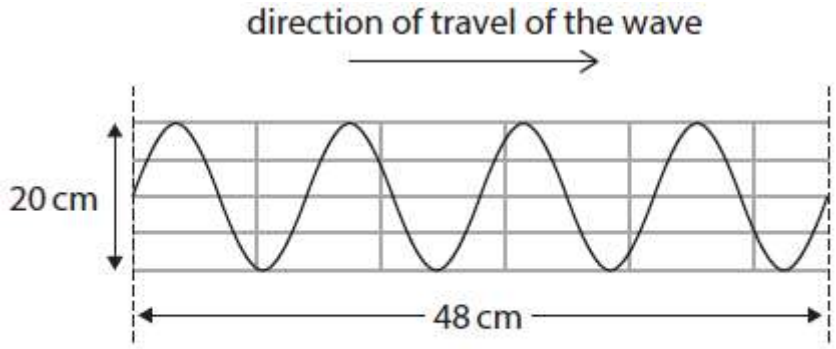

Figure 16

(i) Use data from Figure 16 to determine the amplitude of the wave.

(1)

amplitude = ... cm

(ii) Use data from Figure 16 to determine the wavelength of the wave.

(1)

wavelength = ... cm

(iii) The water wave shown in Figure 16 is generated in a large tank.

A student observes the wave.
Describe how the student could determine the wave speed of the wave.

(2)

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(Total for question = 4 marks)

Q6.

Figure 8 shows a ripple tank.

A screen is placed below the ripple tank.

The wave pattern produced by the ripples can be seen on the screen.

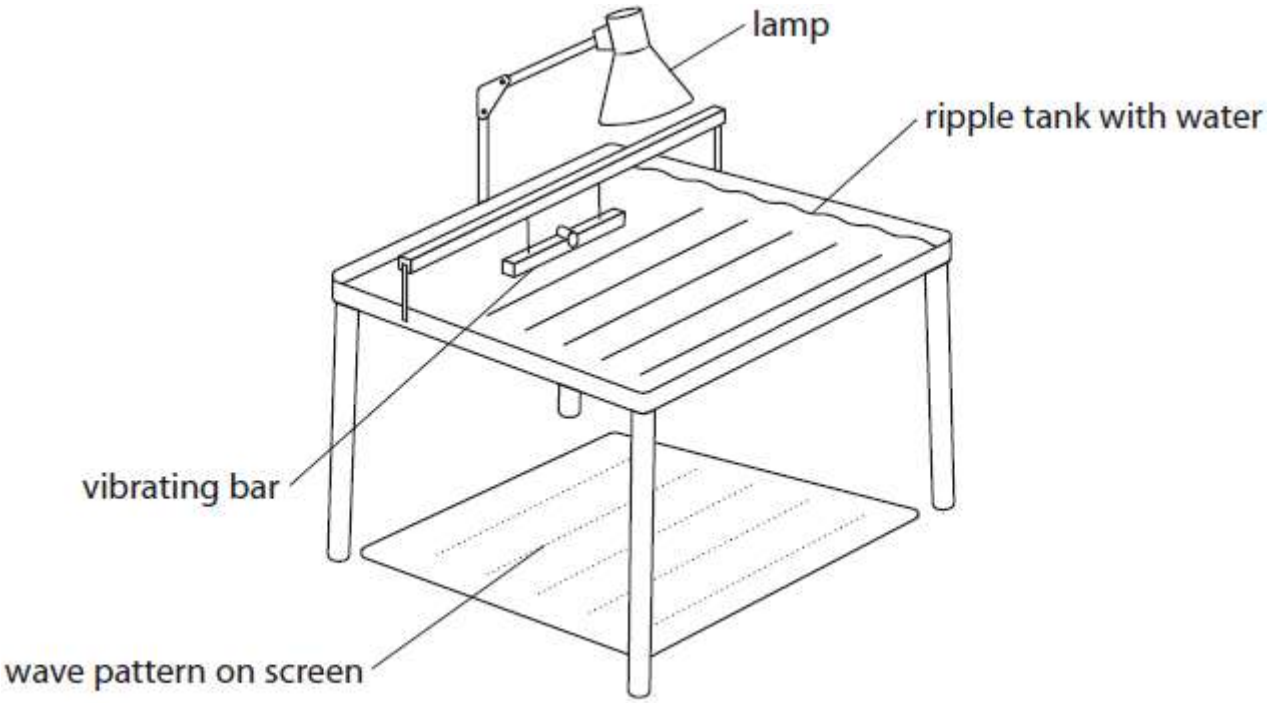

Figure 8

A student has a stop clock and a ruler.

(i) Describe how the student could measure the frequency of the ripples.

(2)

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(ii) Describe how the student could measure the wavelength of the ripples.

(2)

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(Total for question = 4 marks)

Q7.

* A teacher takes a group of students on to a sports field to measure the speed of sound in air.

The teacher fires a starting pistol.
It produces a loud bang and a flash of light at the same time.
The students take measurements.

Describe how the measurements are made and used to determine the speed of sound in air.

You may draw a labelled diagram to help with your answer.

(6)

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(Total for question = 6 marks)

Q8.

Ripples travel out from the centre of a small circular pond to its edge.

(i) Describe how a student could determine the wave speed of the ripples.

(3)

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(ii) Figure 11 shows a duck floating on the pond.

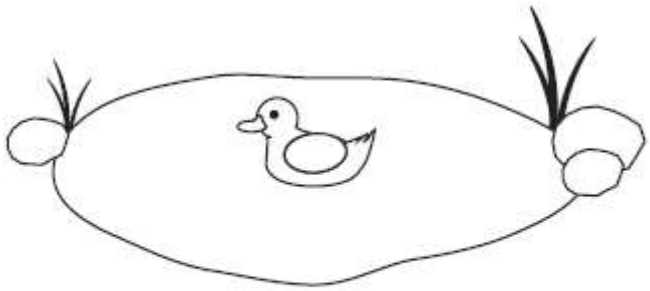

Figure 11

The ripples cause the duck to move.
Draw arrows on Figure 11 to show how the duck moves due to the ripples.

(1)

(Total for question = 4 marks)

Q9.

Some questions must be answered with a cross in a box(☒). If you change your mind about an answer, put a line through the box(☒)and then mark your new answer with a cross(☒).

This question is about waves.

(i) Which row of the table is correct for **sound waves**?

(1)

		sound waves are	can sound waves transfer energy?
☐	A	longitudinal	yes
☐	B	longitudinal	no
☐	C	transverse	yes
☐	D	transverse	no

(ii) A sound wave has a frequency of 440 Hz and a wavelength of 0.75 m.
Calculate the wave speed of the sound wave.

(2)

wave speed = ... m/s

(Total for question = 3 marks)

Q10.

A man throws a stone into a pond.

The stone makes waves that spread out in circles.

Figure 6 shows some of the waves.

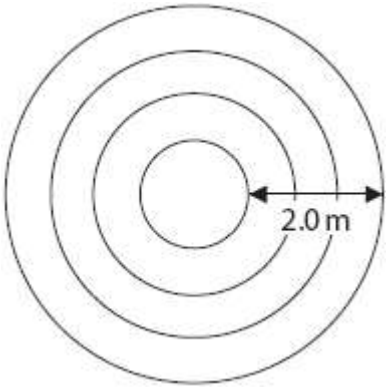

Figure 6

(i) Which of the following changes is correct as the waves spread out?

(1)

☐ **A** the amplitude is higher

☐ **B** the frequency is higher

☐ **C** the wavefront is longer

☐ **D** the period is longer

(ii) The stone hits the water 4.0 m from the bank.

The wave speed is 0.70 m/s.
Calculate the time for the wave to reach the bank.

(2)

time = ... s

(iii) The wavelength of the waves is the distance between one wavefront and the next.

Use the diagram to find the wavelength of the waves.

(1)

wavelength = ... m

(iv) There is a cork which bobs up and down in the water as the wave goes past.

Explain how this shows that the wave is transverse.

(2)

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(Total for question = 6 marks)

Q11.

Figure 11 shows a large tank of water.

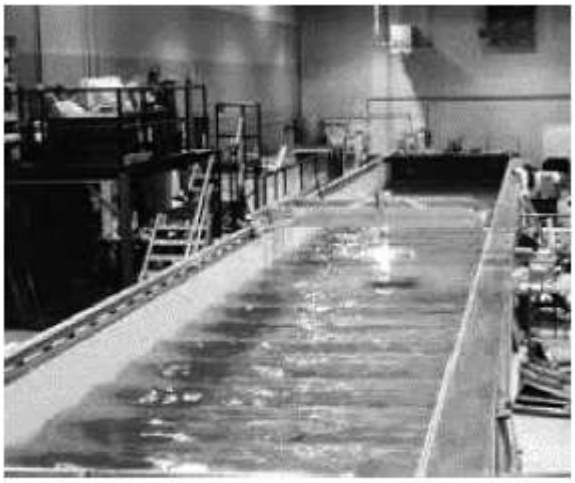

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Figure 11

The tank of water is used to study water waves.

(i) Water waves are transverse waves.
Give another example of a transverse wave.

(1)

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(ii) Figure 12 shows a side view of part of the tank.

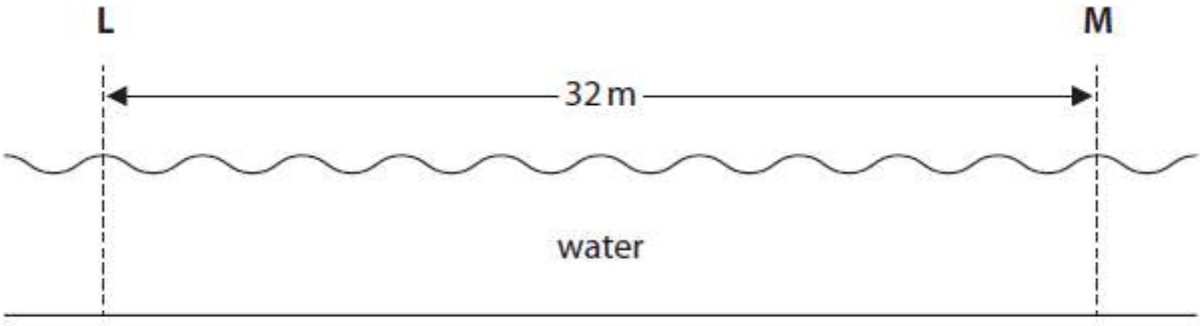


Figure 12

A water wave is moving from **L** to **M**.

Calculate the wavelength of the wave.

(2)

wavelength = ... m

(iii) A technician stands at the side of the tank.

He counts the peaks of the waves as they pass him.
12 peaks pass the technician in a time of 15 s.
Calculate the frequency of the wave.

(2)

frequency = ... Hz

(Total for question = 5 marks)

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
	substitution (1) $\frac{3.0 (\times 10^{8})}{5.8 (\times 10^{-7})}$		(3) AO 2 1
	evaluation (1) 5.2×10^{14}	answers that round to 5.2×10^{14} award 2 marks for a correct answer without working allow 1 mark for answers that round to 5.2 to any power of ten	
	unit (1) Hz	independent mark accept hz or s^{-1} or per sec(ond) or hertz accept kHz, MHz etc with correct power (10^{11} kHz, 10^{8} MHz)	

Q2.

Question number	Answer	Additional guidance	Mark
	an explanation to include two from: waves cannot be seen (on arrival) (1) person will need another way of detecting the waves (1) (as) a person cannot count to 12 in one second / at a rate of 12 per second (1) frequency too high (1)	idea of coming too fast to count / easy to lose count	**(2)**

Q3.

Question Number	Answer	Acceptable answers	Mark
(a)	**A** longitudinal : yes		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)	An explanation linking any two of:		**(2)**
	1. A cause or description of earthquakes (1)	The release of {energy / pressure/friction force} (in Earth's surface) (caused when tectonic) plates slide past each other any idea of relative movement of plates e.g. move over each other, collide	
	2. why timing of earthquake is uncertain / complex (1)	(movement of plates is) {sudden / random / jerky} it is too difficult to {work out / measure} when release of energy will happen	
	3. we cannot see {what is happening deep inside the Earth / where the plates are rubbing} (1)	"it is difficult to measure when the plates will collide" = 2 marks	

Q4.

Question Number	Answer	Additional guidance	Mark
	a description to include:		(3) AO 1 1
	• longitudinal – (vibrations) parallel to (direction of travel) (1)	back and forth (oscillations)/ compressions or rarefactions	
	• transverse – (vibrations) at right angles to (direction of travel) (1)	up and down (oscillations)	
	• (connection between) direction of travel with (direction of) vibrations (1)		

Q5.

Q6.

Question Number	Answer	Additional guidance	Mark
(i)	a description including count the number of waves/ripples (1) (that pass a point) in a certain time (1) OR measure the time for a certain number of waves/ripples (1) use of f = 1/T (1)	accept use of numerical values calculate the number of waves that pass the point in a second scores 2 marks	(2) AO1

Question Number	Answer	Additional guidance	Mark
(ii)	a description including any two from the waves/ripples are made to look stationary (1) measure the distance across a number of waves/wave fronts/ripples (1) calculate the wavelength from the measurements (1)	 using camera, video, strobe light, stroboscope, mobile, phone, photo(graph) accept measure the distance across a number of lines divide distance by the number of waves/ripples accept the idea of measuring the distance between one wave/ripple/line and another (successive) wave/ripple/line for 2 marks	(2) AO1

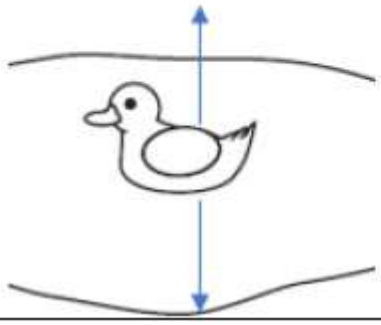
Q7.

Question number	Answer	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative (example) content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 (6 marks)** • large distance between pistol and timer • distance about 300 m • measure distance with trundle wheel / using GPS / mobile phone • measure time between seeing flash and hearing bang • measure time with stop watch • start watch on seeing flash, stop on hearing bang • determine the speed of sound by dividing the distance between the source and the observer by the time difference between the flash and the bang • repeat OR have several students timing the same event • repeat from different directions to take account of wind direction	**(6)**

Level	Mark	Descriptor
	0	No awardable content.
Level 1	1–2	Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3–4	Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5–6	Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Q8.

Question number	Answer	Additional guidance	Mark
(i)	a description to include: use (wave) speed = $\frac{\text{distance}}{\text{time}}$ (1) find relevant time (1) measure specified distance (1)	 use v = f x λ count number of waves in specified time width / radius / circumference of pond do not accept wavelength	**3** **AO2.2**

Question number	Answer	Additional guidance	Mark
(ii)	arrow(s) up and/or down (1) 	judge by eye need not be on duck do not credit answers that imply duck (also) moves horizontally	**1** **AO1.1**

Q9.

Question Number	Answer	Mark
(i)	**A** longitudinal yes ***B*** *is not correct because sound waves can transfer energy* ***C*** *is not correct because sound waves are longitudinal* ***D*** *is not correct because sound waves are longitudinal and sound waves can transfer energy*	**(1)** **AO1**

	Answer	Additional guidance	Mark
(ii)	select wave equation (1) $(v =) f \times \lambda$ evaluation (1) (speed =) 330 (m/s)	 (speed =) freq(uency) × wavelength (speed =) 440 × 0.75 award full marks for the correct answer without working.	**(2)** **AO2**

Q10.

Question number	Answer	Additional guidance	Mark
(i)	C wave front is longer		(1)

Question number	Answer	Additional guidance	Mark
(ii)	substitution and rearrangement (1) t = 4.0 / 0.70 evaluation (1) 5.7 (s)	 accept 6 (s)	(2)

Question number	Answer	Additional guidance	Mark
(iii)	2/3 0.67 m		(1)

Question number	Answer	Additional guidance	Mark
(iv)	An explanation that combines identification - application of knowledge (1 mark) and reasoning/justification - application of understanding (1 mark): the cork is oscillating at right angles / perpendicular (1) to the direction of travel of the wave / transfer of energy(1)		(2)

Q11.

Question number	Answer	Additional guidance	Mark
(i)	**one** from: radio(wave) (1) micro(wave) (1) infrared (1) visible (light) (1) ultraviolet (1) X(-ray) (1) gamma (rays) (1) electromagnetic/em wave(s) seismic **S**(-wave)	Do not credit if sound waves also mentioned γ earthquake **S**(-wave)	**(1)**

Question number	Answer	Additional guidance	Mark
(ii)	$\frac{32}{\text{number of wavelengths}}$ (1) $\frac{32}{10}$ evaluation (1) 3.2 (m)	 accept 9 or 11 for 10 no ecf from mp1 3.6 (3.5r) or 2.9(1) award full marks for the correct answer without working	**(2)**

Question number	Answer	Additional guidance	Mark
(iii)	substitution (1) $\frac{12}{15}$ evaluation (1) 0.8(0) (Hz)	 award full marks for the correct answer without working	**(2)**